

Circuit Breakers

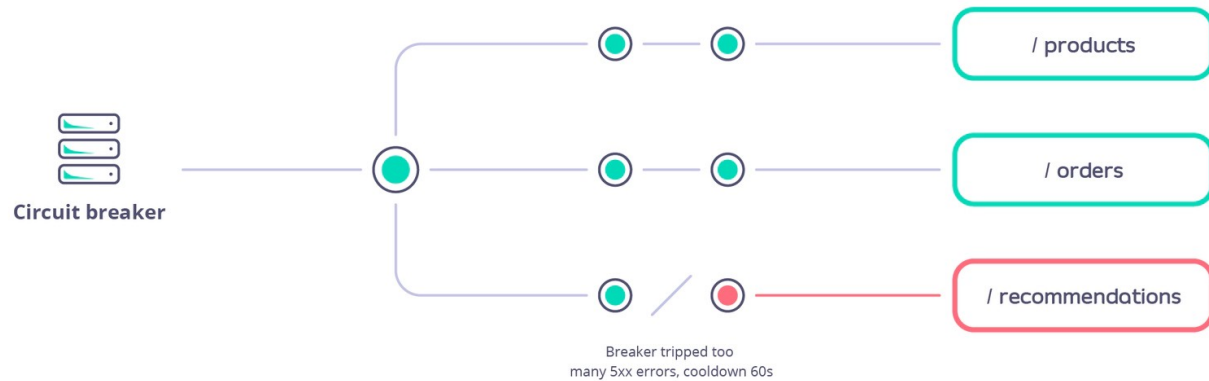
Circuit Breakers

What is a Circuit Breaker?

- A protective mechanism that maintains system stability.
- Prevents repeated failures and overloading of failing services.
- Temporarily blocks calls to unstable services, allowing them time to recover.
- Ensures overall system continues functioning during partial failures.

Circuit Breakers in Tyk

- Configurable per endpoint.
- Monitors HTTP 500+ response rates from upstream services.
- If failures exceed a threshold:
 - Circuit breaker trips.
 - Tyk blocks further requests to that endpoint.
 - Returns HTTP 503 to clients.
- After a defined cooldown period, the breaker resets and resumes traffic.



Circuit Breakers

Event Integration

Tyk can emit events:

- When the circuit breaker trips.
- When it resets.

Use these events for:

- Monitoring & alerting.
- Automation of recovery workflows.

Circuit Breakers

1. Protection of Critical API Endpoints

- Prevents overload of essential APIs
- Ensures uptime and responsiveness under failure scenarios

2. Handling Temporary Issues

- Automatically detects and responds to service degradation
- Opens and closes the circuit based on health, allowing recovery

3. Implementing Retry Logic

- Manages retries after a cooldown period
- Avoids flooding unstable services with repeated calls

4. Enabling Fallback Mechanisms

- Triggers alternate flows when primary services fail
- Supports graceful degradation of service